

Batch: A3 Roll No.: 16010324044
Experiment / assignment / tutorial No. 7
Grade: AA / AB / BB / BC / CC / CD / DD
Signature of the Staff In-charge with date

TITLE: AC analysis of BJT amplifier

AIM: To determine voltage gain (A_v), Resistance (R_i , R_o) of BJT amplifier.

1. **OUTCOME:** Student will be able to analyse transistor amplifier for small signal operation.

Procedure:

Perform the following steps using LTspice and hardware to measure the parameters of an amplifier.

Voltage gain A_v :

Apply the input voltage (V_i) signal in mid frequency range, Measure output voltage (V_o). $A_v = (V_o / V_i)$

Input resistance R_i :

Connect an external source of constant magnitude and frequency (in the mid-band range) at the input through a series resistance R_s . The output voltage is, $V_o = A_v * [R_{in} / (R_s + R_{in})] * V_i$; If $R_s \rightarrow 0$, $V_o \rightarrow A_v * V_i$, and if $R_s = R_{in}$, $V_o = A_v * V_i / 2$.

This gives us the following procedure for finding R_{in} :

1. Vary R_s from 0 (or a negligibly small value) to a large value (10 times the expected value of R_{in}), and plot $|V_o|$ versus R_s .
2. Denote the maximum value of $|V_o|$ by V_m . From the plot, find R_s for which $|V_o|$ is $V_m/2$. The resistance thus obtained is the same as R_{in} .

Output Resistance (R_o):

Apply the input voltage (V_i) signal in mid frequency range. Connect a load resistance R_L at the output. The output voltage is, $V_o = A_v * [R_L / (R_o + R_L)] * V_i$; If $R_L \rightarrow \infty$, $V_o \rightarrow A_v * V_i$ and if $R_L = R_o$, $V_o = A_v * V_i / 2$.

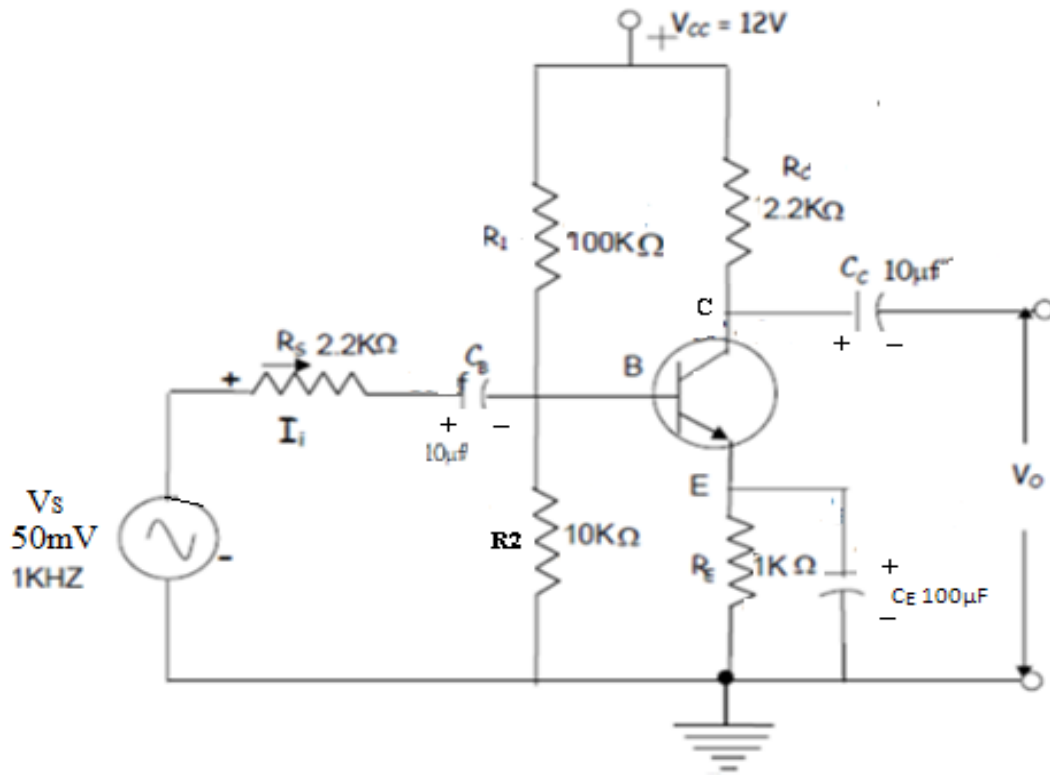
This gives us the following procedure for finding R_o :

1. Vary R_L from ∞ (or a sufficiently large value) to a small value (10 times smaller than the expected value of R_o), and plot $|V_o|$ versus R_L .
2. Denote the maximum value of $|V_o|$ by V_m . From the plot, find R_L for which $|V_o| = V_m/2$. The resistance thus obtained is the same as R_o .

Verify your results with simulation.

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Circuit Diagram:



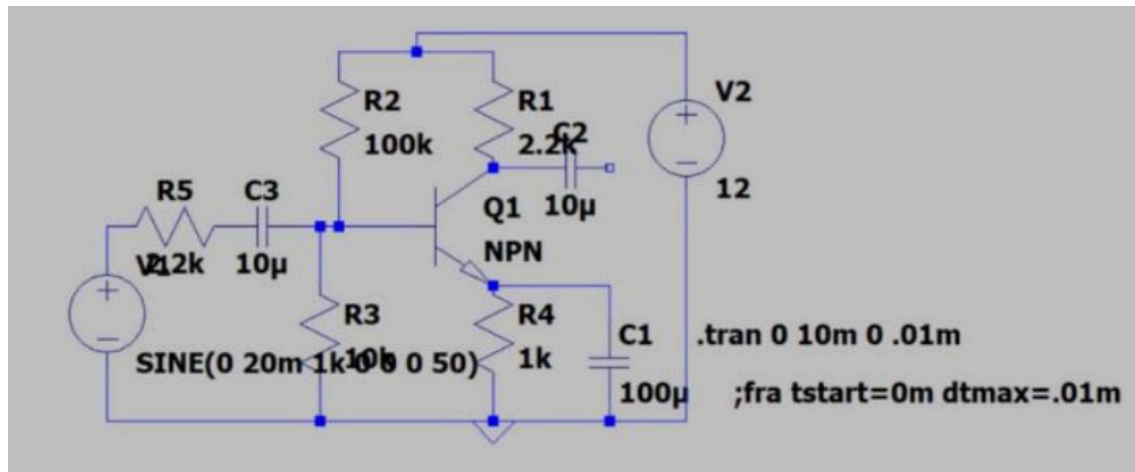
Observation table:

1. Voltage gain of amplifier (A_v)

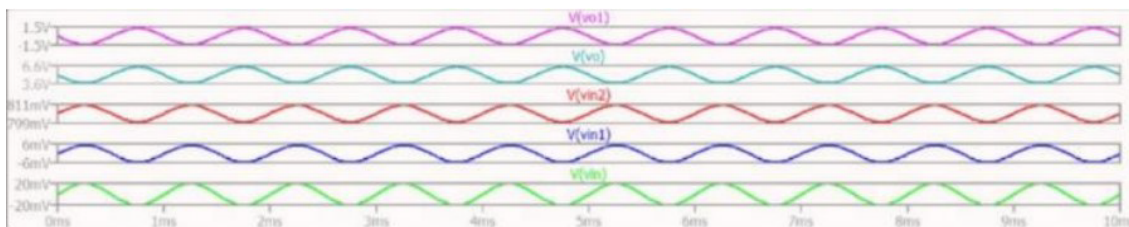
Load Resistor R_L	V_i (mv)	V_o (v)	Voltage gain (A_v) (Practically)	Voltage gain (A_v) (Theoretically)
10V	12V	17.88	1.49	1.49

2. Input Resistance of amplifier (R_i):

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3. Output Resistance of amplifier (R_o):



Calculation:

1. Voltage gain (theoretical) = 1.49
2. Input resistance (theoretical) = 9.09 Kohm
3. Output resistance (theoretical) = 2.2 Kohm

① Voltage Gain (A_v)

$$A_v = \frac{V_{out}}{V_{in}} = \frac{17.88}{12} = 1.4$$

② Input Resistance (R_{in})

$$R_{in} = R_1 || R_2 || (\beta(r_e + R_{EF}))$$

$$r_e = \frac{25 \text{ mV}}{I_E}$$

$$V_B = \frac{V_{CC} \cdot R_2}{R_1 + R_2} = \frac{12 \times 10 \times 10^3}{100 \times 10^3 + 10 \times 10^3}$$

$$V_{BE} = 0.7 \text{ V}$$

$$V_E = V_B - V_{BE}$$

$$= 1.09 - 0.7$$

$$= 0.39 \text{ V}$$

$$I_E = \frac{V_E}{R_E} = \frac{0.39}{1 \times 10^3}$$

$$r_e = \frac{25 \text{ mV}}{I_E} = \frac{25 \text{ mV}}{0.39 \text{ mA}}$$

$$r_e = \frac{25 \times 10^{-3}}{0.39 \times 10^{-3}} = 64.1 \Omega$$

$$r_c = 64.1 \Omega$$

$$\beta(r_e + R_{EF}) = 100 \times (64.1 + 1000)$$

$$= 100 \times 1064.1$$

$$= 106.41 \text{ k}\Omega$$

$$R_{in} = R_1 || R_2 || 106.41 \text{ k}\Omega$$

$$R_1 || R_2 = \frac{R_1 R_2}{R_1 + R_2} = \frac{100 \text{ k}\Omega \times 10 \text{ k}\Omega}{110 \text{ k}\Omega} = 9.09 \text{ k}\Omega$$

$$R_{in} = 9.09 \text{ k}\Omega || 106.41 \text{ k}\Omega = 8.38 \text{ k}\Omega$$

③ $R_{out} \approx \frac{R_c}{1 + \beta}$ Theoretical Value = $2.2 \text{ k}\Omega$

Conclusion:

The AC characteristics of the BJT amplifier were successfully studied. The amplifier showed a stable voltage gain with predictable input and output resistances, confirming its suitability for signal amplification applications.

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